



Trend-weighted multi-resolution transformer for multi-parametric glucose prediction

Ömer Atılım Koca^a, Volkan Kılıç^b,*

^a Department of Software Engineering Graduate Program, İzmir Katip Celebi University, İzmir, Turkey

^b Department of Electrical and Electronics Engineering, İzmir Katip Celebi University, İzmir, Turkey

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ABSTRACT

Accurate glucose level prediction using continuous glucose monitoring data is crucial for effective diabetes management, enabling timely interventions to reduce hypo- and hyperglycaemic events. However, predicting future glucose levels remains a challenging due to the complex and variable nature of glucose dynamics. While artificial intelligence-based models have been widely applied to address these complexities, traditional approaches often lack the ability to capture the temporal dependencies. To address this limitation, we propose the Trend-Weighted Multi-Resolution Transformer (TW-MRT) model that effectively captures the complex temporal dependencies in glucose dynamics. TW-MRT introduces a trend-weighting mechanism to adaptively focus on relevant glucose trends, while a multi-resolution fusion layer processes data across varying time scales. This integrated layer allows the model to dynamically adjust glucose patterns, enhancing its predictive accuracy. The proposed model was evaluated on both the OhioT1DM and D1NAMO datasets, achieving root mean square error value of 11.37 mg/dL on OhioT1DM, and achieved an average RMSE of 11.30 mg/dL on D1NAMO for 30 PH. These results demonstrate the superior performance of the proposed TW-MRT model compared to existing state-of-the-art models. Furthermore, an ablation study confirms the contribution of each component within the model, demonstrating the importance of trend-weighting mechanism and multi-resolution fusion in improving predictive performance. Key contributions include: (1) trend-weighting and multi-resolution fusion enabling capture of short-, mid-, and long-term glucose dynamics, (2) cross-dataset validation demonstrating robust generalizability across diverse populations and sensor technologies, and (3) comprehensive clinical validation through surveillance error grid analysis, which confirms clinically reliable predictions across all patient scenarios.

1. Introduction

The prediction of glucose levels is essential in diabetes management, as it provides sufficient time to implement preventive strategies, helping to maintain stable glucose levels and reduce the risk of diabetes-related complications [1]. Recent advancements in technology have enabled the development of continuous glucose monitoring (CGM) devices, which deliver sequential and time-varying glucose readings at frequent intervals, typically every five minutes. These devices provide a more detailed representation of glucose trends compared to conventional methods such as fingerstick testing and laboratory measurements [2]. The analysis of these trends enables the identification of temporal patterns and the complexity of glucose fluctuations, which can be used for prediction. However, glucose prediction remains a challenging task due to the highly complex and nonlinear nature of its fluctuations, often resulting in limited precision.

In recent years, artificial intelligence (AI)-based models have emerged as a useful tool to address these challenges [3]. These models, particularly recurrent neural networks (RNNs) and their variants, have proven effective in handling sequential data and identifying temporal patterns [1]. However, CGM data exhibit complex temporal dependencies influenced by factors such as meals, physical activity, and insulin usage, which vary across different time periods. This complexity presents a challenge for traditional models, as they often lack the ability to fully capture these dependencies, resulting in limited capacity to manage long-term patterns while identifying trends and localized fluctuations in glucose levels.

In this regard, the emergence of transformer-based models has represented an advance in time series analysis, offering a novel alternative to RNNs with their self-attention mechanisms [4]. These transformer-based models have shown a robust ability to capture complex temporal relationships and effectively analyze patterns within sequential glucose

* Corresponding author.

E-mail address: volkan.kilic@ikcu.edu.tr (V. Kılıç).

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data [5]. However, current transformer-based models are unable to effectively capture glucose fluctuations at different scales, including detailed local changes and broader overall trends, which are essential for accurate glucose prediction. This paper proposes a novel Trend-Weighted Multi-Resolution Transformer (TW-MRT) model, which has been specifically designed to address the challenges of glucose prediction. Our model combines the strengths of transformer architectures with a trend-weighting and multi-resolution fusion, to effectively capture both short-term fluctuations and long-term trends in glucose levels. Trend-weighting mechanism is designed to adaptively focus on significant glucose patterns and differentiate between meaningful trends and short-term variations. This mechanism enables the model to prioritize relevant trends, allowing it to effectively balance short-term and long-term pattern recognition, thereby improving predictive accuracy across various time scales. In addition to the trend-weighting mechanism, the model employs a multi-resolution fusion layer that processes data at different temporal scales. This multi-scale analysis allows the model to capture both rapid changes in glucose levels and slower, long-term trends, which enables a comprehensive view of glucose dynamics [6]. Furthermore, the TW-MRT model incorporates a feature importance layer, which is an adaptive system that adjusts the weights of input features based on their importance at different times. This layer enhances the capacity of the model to make accurate predictions through the focus on the most relevant features. Finally, the TW-MRT model includes a time-aware embedding system to allow the incorporation of temporal data directly into the input dataset. The system allows the model to capture the time-dependent nature of glucose fluctuations by adjusting the importance of features [7]. Furthermore, we employed a multi-parametric approach by incorporating basal and bolus insulin as input features to improve the predictive performance of the proposed TW-MRT model. This multi-parametric approach enables the model to analyze the impact of insulin dynamics on glucose levels, which is crucial for capturing the response of the body to both basal insulin doses and the rapid fluctuations caused by bolus insulin injections. This multi-parametric approach, combined with the trend-weighting and multi-resolution analysis, enables the TW-MRT model to provide a comprehensive and adaptive understanding of glucose dynamics, which is particularly beneficial for real-world diabetes management scenarios. The major contributions of this paper can be summarized as follows:

- The introduction of a trend-weighting mechanism and multi-resolution fusion will improve the ability of the model to identify meaningful glucose trends and short-term fluctuations across a range of time scales.
- The development of a TW-MRT model will allow the capture of both global and local patterns in glucose time-series data, while enabling adaptive adjustment of the relative importance of different input features.
- The performance of the proposed TW-MRT model was evaluated against state-of-the-art glucose prediction methods on multiple prediction horizons (PH), demonstrating its effectiveness in managing the complex dynamics of blood glucose levels.

The rest of this paper is organized as follows: Section 2 reviews related work on glucose prediction and discusses the limitations of existing methods. Section 3 outlines the background and methodology of the proposed TW-MRT model. The dataset, implementation details, and comparison with state-of-the-art models are presented in Section 4. Finally, concluding remarks and future research directions are provided in Section 5.

2. Related works

This section describes the evolution of glucose prediction, deep learning (DL) models in glucose prediction, the application of transformer models in time series analysis, and the importance of the trend-weighting mechanism and multi-resolution analysis, aiming to provide the background for understanding the context and motivation behind the proposed TW-MRT model.

2.1. Evolution of glucose prediction models

Glucose prediction has become an important research area due to its essential role in improving diabetes management and reducing the risk of hyperglycemic or hypoglycemic events. Traditional models for the prediction of glucose levels relied mainly on mathematical models and statistical techniques, such as autoregressive models and linear regression analysis [8]. Among these, physiological models such as the Bergman minimal model and polynomial models have attempted to characterize the glucose-insulin system through differential equations [9,10]. In addition, time series models including autoregressive integrated moving average (ARIMA) and autoregressive moving average with exogenous inputs (ARMAX) models were used for short-term glucose prediction. Prendin et al. developed ARIMA models for glucose prediction using CGM data provided by Dexcom, demonstrating the applicability of statistical time series techniques to glucose prediction [11]. Turksoy et al. introduced ARMAX-based models for early hypoglycemia alarms [12]. While these models provided valuable insights, their predictive capacity was limited by their inability to capture non-linear relationships in glucose dynamics.

Advancements in machine learning (ML), subset of AI, have led to the development of novel predictive models that show enhanced ability to capture the complexities in glucose patterns. Traditional ML-based models, such as Support Vector Machines and Random Forests, show improved performance due to their ability to process and analyze multiple input features, such as insulin doses, carbohydrate intake, and physical activity. Hamdi et al. developed support vector regression models integrated with differential evolution model to enhance the accuracy of glucose prediction [13]. Georga et al. proposed a Random Forest-based model for predicting subcutaneous glucose concentrations using physiological and behavioral features from the METABO dataset [14]. Additional ML techniques like Gaussian Process Regression and Gradient Boosting Trees have been applied to glucose prediction, showing promising results in capturing complex patterns in physiological data [15]. Zaidi et al. developed Gaussian Process models for multi-step-ahead glucose prediction using data from the Tidepool platform [16], while Alfian et al. proposed the use of XGBoost models to predict glucose levels on the DirecNet dataset [17]. Saiti et al. proposed an ensemble-based model that integrates bagging, boosting, and linear meta-regressors with a compartmental model of insulin dynamics, aiming to enhance the robustness and adaptability of glucose prediction [18]. Basile and Sannino conducted a comparative analysis of interpretable ML models for glucose prediction, including Bayesian Ridge Regression, decision trees, and boosting-based algorithms. Their findings demonstrate the effectiveness of interpretable models in clinical time series prediction, with Bayesian Ridge achieving the best average performance across patients [19]. However, the complexities of handling time dependencies and multi-parameter data, such as capturing non-linear relationships and learning complex temporal patterns, have highlighted the necessity for more sophisticated models. This has led to the incorporation of DL, which has the capability to extract meaningful features and modeling dynamic interactions within high-dimensional data [20].

2.2. Deep learning models in glucose prediction

The application of DL models has led to an advancement in glucose prediction, with the ability to capture complex, non-linear temporal relationships within the dataset. RNNs and their variants, particularly long short-term memory (LSTM) networks, have been widely utilized for glucose prediction. Sun et al. proposed LSTM and bidirectional LSTM models for glucose prediction, using deep neural networks to model temporal dependencies in the GoCARB dataset [21]. Nemat et al. proposed integrating ML and DL algorithms for glucose prediction, achieving a performance improvement compared to traditional ML methods [22]. However, the utilization of single-layer

architectures limited the ability of the model to effectively capture the complex dependencies present in glucose data. The limitations of single-layer networks become particularly evident in modeling the multiscale temporal dynamics of glucose fluctuations, which include both rapid changes and slower trends throughout the day. These models often suffer from the vanishing gradient problem, which inhibits their ability to learn long-term dependencies critical for accurate glucose prediction [23]. In order to address these limitations, more sophisticated architectures were explored, including stacked neural networks and models that incorporate advanced mechanisms for managing sequential data [8]. Stacked models, which employ multiple layers of neural networks, enable the model to explore the temporal patterns and relationships of the data in depth [24]. Zhu et al. proposed DL models that integrate the Internet of Medical Things (IoMT) with RNN, using multiple input windows to represent different temporal patterns [25]. This model leverages data from wearable devices and CGM to create a comprehensive framework that adapts to individual patient characteristics. Rabby et al. proposed a deep RNN model based on Kalman smoothed stacked LSTM for glucose prediction [8]. The objective of this neural network was to enhance prediction accuracy through the utilization of Kalman smoothing and stacked layers. This combination of statistical filtering techniques with DL architectures highlights the increasing focus on hybrid models, which aim to integrate the complementary strengths of different modeling techniques. Other hybrid DL methods include the weighted decision-level fusion model, which combines outputs from multiple neural predictors through fusion strategies to enhance robustness [26], and ensemble neural networks that aggregate predictions from various LSTM-based learners [27]. GluNet employs a personalized deep architecture that processes multi-parametric patient histories for individualized prediction [28], while the autonomous-channel DL model uses parallel asynchronous LSTM channels that self-select relevant input lags [29]. Generative frameworks such as GAN with GRU aim to refine predictions through adversarial learning [30], and deep reinforcement learning models optimize LSTM weights via reward-based updates to improve prediction over long horizons [31]. Gu et al. introduced a neural physiological encoder that decomposes physiological signals using CNN-based filters [32], while Butt et al. proposed a Bi-LSTM framework enhanced by feature transformation techniques [3]. The GLIMMER model combines CNN and LSTM modules with a custom error-rate loss to enhance multistep prediction performance [33]. Zhu et al. developed models using Edge-LSTM and Dilated Recurrent Neural Networks (DRNN) to handle features across different temporal resolutions, enhancing the prediction accuracy by managing long-term dependencies [34,35]. Although these stacked architectures demonstrated some advancements in prediction capability, they frequently encountered challenges in capturing the glucose trend, especially during the analysis of varying time scales. These limitations have motivated the development of more adaptive models capable of focusing on both short-term changes and long-term trends.

2.3. Transformer models and attention mechanisms in time series analysis

The transformer model and its attention mechanisms have revolutionized time series analysis, enabling models to focus on different parts of the input sequence, and to capture of both short and long-term dependencies. Traditional RNNs, LSTMs, and GRUs were often constrained by their sequential processing nature, whereas transformers process the entire sequence in parallel, leading to better performance, especially in complex time series tasks [39]. The transformer, originally developed for natural language processing tasks, has been successfully adapted for time series prediction due to its self-attention mechanism [40]. This mechanism allows the model to dynamically weight the importance of different time steps in the input sequence, providing a more flexible model to capturing temporal dependencies than the fixed memory mechanisms of RNNs [41]. The parallel computational capability of transformers also overcomes the computational limitations

associated with sequential processing in RNNs, enabling more efficient training across diverse time series applications, including glucose datasets. The Crossformer introduces a two-stage attention mechanism that separates temporal and dimensional attention with patch-wise processing, enabling more efficient modeling of multivariate time series [42]. The edge-based temporal fusion transformer (E-TFT) adapts temporal fusion architectures for edge computing, using simplified attention mechanisms suited for deployment in resource-constrained environments [43]. The population-specific TFT model integrates demographic features into transformer-based glucose prediction, aiming for improved personalization across different patient populations [44]. Yang et al. explored the multi-head attention networks for glucose prediction, showing that attention mechanisms enhance short-term predictions [38]. These multi-head attention networks divide the attention computation into multiple heads, enabling the model to focus on different subspaces of representation and temporal patterns in parallel. In our previous study [36], we proposed a DL model that combined a stacked GRU encoder with an attention-based GRU decoder to improve the modeling of temporal dependencies in glucose time series. This model aimed to dynamically emphasize informative time steps during decoding, enhancing short-term prediction performance. Zhu et al. proposed a meta-learning model for glucose prediction that uses attention layers to effectively capture complex patterns in glucose time series [37]. Although this model can be effective for certain time windows, it may fail to capture the dynamic nature of glucose trends over different time scales. Through focusing on fixed window inputs, these models can miss critical variations that occur at different temporal resolutions. To address these issues, the proposed TW-MRT aims to provide a more comprehensive and flexible model for glucose prediction by adaptively weighting significant trends and processing data across multiple temporal scales.

2.4. Trend-weighting mechanism and multi-resolution analysis in predictive modeling

The adaptive capacity of models to the complex and fluctuating nature of glucose dynamics enhances their performance in glucose prediction. Traditionally, glucose prediction models have relied on fixed input features and single-scale analysis, limiting their ability to capture varying temporal patterns effectively [45]. To address this limitation, multi-resolution analysis in time series prediction has been explored and demonstrated its potential to process information at different time scales. Multi-resolution analysis, a technique commonly employed in DL models for time series tasks, allows for the processing of information across multiple time scales, enabling the capture of both short-term fluctuations and longer-term trends [46]. Zhang et al. employ a multi-resolution-based model, known as frequency enhanced decomposed transformer, for long-term time series prediction [47]. This model decomposes input time series into multiple frequencies, which enables the capture of complex patterns and enhances prediction accuracy. However, while these multi-resolution approaches have shown promise in general time series prediction, their application to glucose prediction remains limited due to the unique and nonlinear dynamics of glucose levels, which are influenced by factors such as physiological variability and external factors such as meals and insulin. Most glucose prediction models continue to depend on fixed input features and single-scale analysis, which limits their adaptability to the diverse temporal patterns in glucose data [11]. Here, our proposed TW-MRT model introduces a trend-weighting mechanism specifically designed for glucose prediction. Unlike existing models, this mechanism dynamically assigns weights to different glucose trends, allowing the model to emphasize important patterns as they develop over time. This focus on the most relevant changes in the glucose time series improves the ability of the model to make accurate predictions, particularly in capturing the challenging fluctuations in glucose levels. Furthermore, the integration of trend-weighting with multi-resolution analysis in the TW-MRT model

Table 1
Performance of glucose prediction models on various datasets (RMSE in mg/dL).

Ref	Model	Dataset	RMSE
[19]	Bayesian Ridge	D1NAMO	1.59 (mmol/L)
[14]	Random Forest	METABO	8.15
[13]	SVR + Differential Evolution	CGM data, 12 T1D patients	10.78
[36]	Attention-based multilayer GRU	OhioT1DM dataset	13.11
[3]	Feature Transformation Bi-LSTM	OhioT1DM dataset	16.81
[12]	ARMAX	University of Illinois at Chicago	17.47
[18]	BAGGING MetaR	D1NAMO	17.66
[32]	Neural Physiological Encoder	OhioT1DM dataset	17.80
[31]	Deep RL with LSTM	OhioT1DM dataset	18.32
[30]	GAN with GRU	OhioT1DM dataset	18.37
[8]	Stacked LSTM + Kalman	OhioT1DM dataset	18.57
[37]	Evidential DL with meta-learning	OhioT1DM dataset	18.64
[34]	Dilated RNN (DRNN)	OhioT1DM dataset	18.70
[29]	Autonomous-channel DL	OhioT1DM dataset	18.93
[35]	Edge-LSTM	OhioT1DM dataset	19.10
[28]	GluNet	OhioT1DM dataset	19.28
[27]	Ensemble Neural Networks	OhioT1DM dataset	19.57
[22]	Deep Ensemble Learning	OhioT1DM dataset	19.62
[38]	Multi-head Attention	OhioT1DM dataset	20.57
[21]	LSTM/BiLSTM	GoCARB	21.75
[26]	Weighted Decision-Level Fusion	OhioT1DM dataset	21.90
[11]	ARIMA	CGM data provided by Dexcom	22.15
[17]	XGBoost	DirecNet	23.22
[33]	GLIMMER	OhioT1DM dataset	23.97
[16]	Gaussian Process	Tidepool	28.01

allows for the processing of data at multiple resolutions, leading to a more robust and adaptable prediction framework. The proposed model enables the capture of glucose dynamics across a range of time scales, together with the adjustment of focus in alignment with significant trends. The incorporation of these novel components enhances the TW-MRT model and provides a more comprehensive solution to the complexities of glucose prediction.

To summarize, glucose prediction has evolved from statistical and ML models to advanced DL models, each progressively addressing the nonlinear and temporal complexities of glucose dynamics. Early models such as ARIMA, ARMAX, and tree-based ensemble methods provided initial baselines, while deep neural networks — including LSTM, Bi-LSTM, and GRU-based models — enabled the extraction of richer temporal features. Transformer-based architectures have recently emerged as powerful tools for modeling sequential data, offering greater flexibility in capturing long- and short-term dependencies through attention mechanisms. Furthermore, recent research emphasizes trend-weighting and multi-resolution analysis as critical components for adapting to the diverse temporal patterns of glucose fluctuations. This paper is an enhanced version of our previous study described in [36], expanding upon the attention-based GRU framework by introducing a novel trend-weighted, multi-resolution temporal model. The performance comparisons of existing models are summarized in Table 1 to provide a comprehensive overview of the current state-of-the-art in glucose prediction.

3. Method

This section presents the proposed TW-MRT model, which has been developed to address the complexities of glucose prediction through data preprocessing techniques and an innovative DL model. The proposed model covers data preparation steps, including imputation, feature selection, normalization, and smoothing, to the design and implementation of the TW-MRT model, which integrates a feature importance layer, time-aware embedding, trend-weighting mechanism, multi-resolution analysis, and transformer.

3.1. Data preprocessing

In time series prediction, preprocessing is crucial for enhancing pattern recognition by mitigating noise, resolving inconsistencies, and

improving the capacity of the model to capture temporal dynamics, thereby ensuring accurate predictions [48]. The raw time series data often presents several issues that can impact the performance of the model, including missing values, inconsistent scales, and noise that may affect the visibility of key patterns [49,50]. To address these issues, several techniques such as data imputation, parameter selection, normalization, and smoothing are employed in the preprocessing steps to transform raw time series into a structured, high-quality dataset essential for optimizing model performance and accuracy [51].

The data imputation is a technique utilized to fill gaps caused by the missing values. A common technique for filling these gaps is linear interpolation, which estimates the missing values based on the assumption of a linear progression between two known data points [52]. Additionally, forward and backward filling are two commonly used imputation techniques. These techniques propagate known values to fill adjacent gaps, ensuring continuity without introducing discontinuities. Forward filling uses the previous value, while backward filling uses the next one. Maintaining temporal sequence is crucial in time series analysis, as missing data can obscure patterns and reduce the capacity of the model to learn temporal dependencies.

Parameter selection is a critical preprocessing step in time series prediction, aiming to refine the dataset by identifying the most relevant variables for the predictive task. A common approach for parameter selection is the correlation heatmap, which provides a visual representation of the relationships between variables [53]. These heatmaps are constructed using correlation coefficients (e.g., Pearson's r), which quantify the strength and direction of the relationship between each feature and the output. The coefficient ranges from -1 to 1 , where values close to -1 or 1 indicate strong negative or positive correlation, respectively, and values near 0 indicate weak or no linear relationship. The heatmap guides the selection of features with the highest predictive value, ensuring that the model focuses on informative inputs while minimizing the influence of irrelevant or weakly correlated parameters.

On the other hand, time series data often includes features in diverse ranges, and without scaling, features with larger magnitudes may dominate the learning process of the model [54]. Standard scaling, which adjusts each feature to have a mean of zero and a standard deviation of one, ensures that all features contribute equally during training.

Kalman smoothing is a commonly used technique for reducing noise in time series data, aiming to reveal the underlying trends more

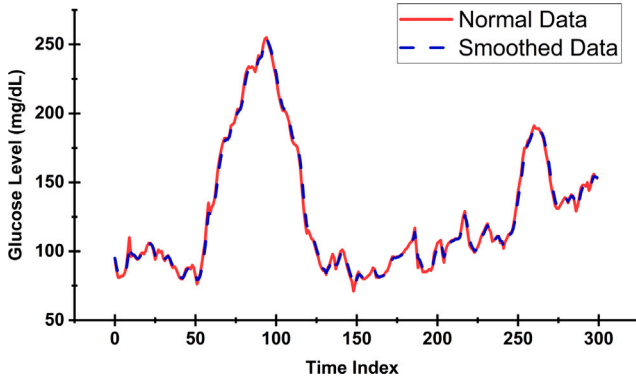


Fig. 1. Actual data vs smoothed data with UKS for patient 552.

accurately [55]. The unscented Kalman smoothing (UKS) method is well-suited for analyzing time series with complex, non-linear behaviors and varying patterns. UKS improves data quality by predicting hidden states and filtering of the series, which removes short-term fluctuations that may hide important trends. Unlike the traditional Kalman filter, which relies on linearization, the unscented Kalman filter uses the unscented transform to propagate the mean and covariance through nonlinear functions without making a linear approximation. Specifically, UKS uses scaled sigma points, as introduced by van der Merwe. These are deterministically chosen samples that capture the distribution of the input state. These sigma points are passed through the nonlinear system, and their weighted outcomes are used to reconstruct the posterior mean and covariance more accurately. Compared to simpler methods such as linear interpolation or forward-filling, UKS offers the additional benefit of uncertainty quantification, which is essential in clinical settings. Previous studies have highlighted the effectiveness of Kalman smoothing in glucose data processing. Rabby et al. reported that applying smoothing to CGM signals improves the accuracy of DL-based glucose prediction [8]. Similarly, Staal et al. demonstrated that Kalman smoothing performs better than traditional filtering or interpolation methods by identifying and correcting inconsistencies in sensor measurements [56]. Fig. 1 illustrates the effect of UKS on noisy data, resulting in a more accurate and reliable dataset. This step is essential for improving the ability of the model to capture the true dynamics of the data, which enhances the predictive performance.

3.2. Proposed model: trend-weighted multi-resolution transformer

The proposed TW-MRT model is designed to address time-series prediction challenges by incorporating four key components: feature weighting, time-aware embeddings, glucose trend-weighting, and a multi-resolution transformer. This integration aims to benefit from complementary characteristics of each components, highlighting respective strengths and addressing weaknesses to improve predictions.

Fig. 2 illustrates the overall schematic of the proposed TW-MRT model capable of capturing both local and global data patterns. The components in the model is designed to prioritize the most important features, capture relevant temporal patterns, apply trend-weighting to focus dynamically on short-term, mid-term, and long-term glucose trends, and leverage multi-resolution fusion to extract information across various temporal scales.

Each component has specific tasks for processing and refining the data before making the final prediction, and their interactions enables the model to capture both short and long-term fluctuations in glucose dynamics while maintaining robust predictive accuracy across multiple time horizons. The model utilizes a sliding window approach that enables sequential data to be processed continuously while maintaining temporal relationships between data points to generate the input data

Algorithm 1 Proposed TW-MRT Model

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1: Input: Time series  $\mathbf{X} \in \mathbb{R}^{N \times F}$ , ground truth  $\mathbf{y} \in \mathbb{R}^N$ , epochs  $E$ , learning rate  $\eta$ 
2: Output: Optimized model parameters  $\theta$ 
3: Initialize  $\theta = \{\mathbf{W}_{\text{imp}}, \mathbf{W}_{\text{proj}}, \mathbf{W}_{\text{trend}}, \mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V, \mathbf{W}_{\text{conv},r}, \mathbf{W}_{\text{mr}}, \mathbf{W}_{\text{out}}\}$ 
4: for epoch = 1 to  $E$  do
5:   Compute feature importance:  $\mathbf{F} = \mathbf{X} \odot \mathbf{W}_{\text{imp}}$ 
6:   Apply time embedding:  $\mathbf{E} = (\mathbf{F}\mathbf{W}_{\text{proj}}) \oplus \text{TimeEmbed}$ 
7:   Trend weighting:  $\mathbf{E}_W = \text{TrendWeighting}(\mathbf{E}, \mathbf{W}_{\text{trend}})$ 
8:   for  $l = 1$  to  $L$  do //  $L$  is the number of transformer layers
9:      $\mathbf{Q} = \mathbf{E}_W \mathbf{W}_Q, \mathbf{K} = \mathbf{E}_W \mathbf{W}_K, \mathbf{V} = \mathbf{E}_W \mathbf{W}_V$ 
10:     $\mathbf{A} = \text{softmax}((\mathbf{Q}\mathbf{K}^T)/\sqrt{d_k})$ 
11:     $\mathbf{Z}_{\text{attn}} = \mathbf{A}\mathbf{V}$ 
12:     $\mathbf{Z}_{\text{out}} = \text{LayerNorm}(\mathbf{Z}_{\text{attn}} \oplus \text{FeedForward}(\mathbf{Z}_{\text{attn}}))$ 
13:   end for
14:   Initialize multi-resolution features:  $\mathcal{O}$ 
15:   for  $r = 1$  to  $R$  do //  $R$  is the number of resolutions
16:      $\text{conv\_out} = \sum_{n=1}^N \mathbf{Z}_{\text{out}}[n : n + 2^r, :] \odot \mathbf{W}_{\text{conv},r}$ 
17:      $\mathcal{O} = \left[ \mathcal{O} \quad \frac{1}{N} \sum_{n=1}^N \text{conv\_out}[n, :] \right]$ 
18:   end for
19:   Concatenate and fuse:  $\mathbf{O} = \text{Concatenate}(\mathcal{O})\mathbf{W}_{\text{mr}}$ 
20:   Generate prediction:  $\hat{\mathbf{y}} = \mathbf{O}\mathbf{W}_{\text{out}}$ 
21:   Compute loss:  $\mathcal{L} = \text{MSE}(\hat{\mathbf{y}}, \mathbf{y})$ 
22:   Update parameters:  $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$ 
23: end for
24: return  $\theta$ 

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of TW-MRT model. At each time step t , the input consists of glucose G , basal insulin I_{Bo} , and bolus insulin I_{Bo} . The model processes these inputs over a sliding window of n previous time steps, denoted as $[t - n, t]$, enabling it to capture historical trends and temporal dependencies for future predictions. The model is capable of predicting future glucose levels based on data sampled at 5 min intervals. This enables the model to make predictions for a range of PH, including 15, 30, 45, 60, 90, and 120 min.

The TW-MRT model uses a multi-faceted approach to capture the complex relationships between features and glucose dynamics across a range of time scales. This is achieved through the integration of feature importance, time-aware embeddings, glucose trend-weighting, transformer, and multi-resolution fusion. This hybrid approach enables the model to prioritize key features, adapt to temporal contexts, and focus on relevant glucose trends, resulting in improved prediction accuracy.

Although similar models have been used these components in other time-series applications, their integration in this work is customized to the unique characteristics of glucose dynamics. First, our trend-weighting mechanism is physiologically grounded, targeting three glucose dynamics: (1) acute insulin response trends (0–30 min) capturing immediate bolus insulin effects, (2) intermediate glucose fluctuation trends (30–90 min) modeling complex insulin interactions, and (3) extended glucose stability trends (90–120 min) tracking longer-term basal insulin effects. This physiological division allows for adaptive weighting based on the physiology of diabetes rather than generic temporal patterns. Second, multi-parametric processing addresses the temporal lag between insulin and glucose, which is missing in standard transformers. Our model incorporates feature-specific temporal embeddings that consider the delayed glucose response to bolus insulin (peaking 15–45 min after administration) and basal insulin (onset after 1–2 h). This ensures that the model captures the causal relationship between insulin dosing and glucose changes. Third, our multi-resolution fusion targets clinically relevant glucose monitoring scales, including short-term fluctuations (five to 15 min), intermediate dynamics (15 to 60 min), and long-term stability patterns

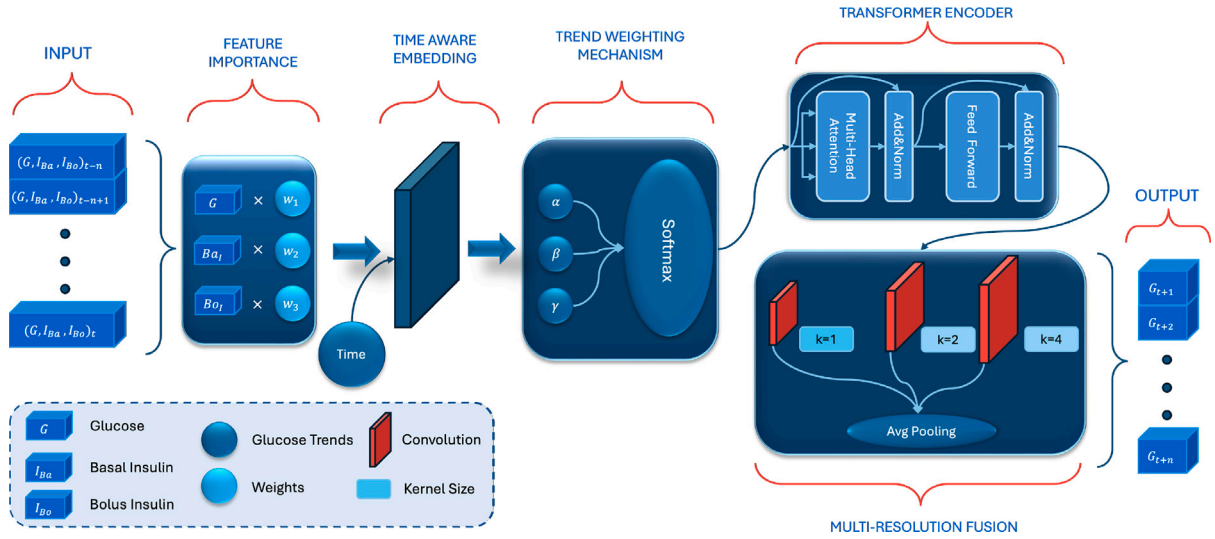


Fig. 2. The proposed TW-MRT Model.

(60 to 120 min), aligning multi-scale analysis with diabetes management timescales. Unlike existing transformer-based glucose prediction models that apply generic attention mechanisms, our model integrates these physiologically-motivated components within a unified framework specifically designed for glucose–insulin dynamics, enabling adaptive focus based on current glucose state and insulin activity rather than static attention patterns. The overall training process of this hybrid model is summarized in Algorithm 1, where each dataset passes through the layers sequentially and is optimized according to the model loss function. In this algorithm, $\mathbf{X} \in \mathbb{R}^{N \times F}$ represents the input time series with N time steps and F features, and $\mathbf{y} \in \mathbb{R}^N$ is the ground truth glucose values. The model parameters θ consist of several weight matrices: the feature importance matrix $\mathbf{W}_{\text{imp}}$, the time embedding projection matrix $\mathbf{W}_{\text{proj}}$, and the trend weight matrix $\mathbf{W}_{\text{trend}}$. The transformer attention mechanism uses the matrices $\mathbf{W}_Q$, $\mathbf{W}_K$ and $\mathbf{W}_V$ to generate the query (Q), key (K) and value (V) representations respectively. The attention weights are represented by A, which are computed from the query and key representations. The intermediate features produced by the attention mechanism are denoted as $\mathbf{Z}_{\text{attn}}$, while $\mathbf{Z}_{\text{out}}$ represents the final encoded sequence after the transformer block. In addition, the model uses convolution matrices $\mathbf{W}_{\text{conv},r}$ for multi-resolution analysis, along with fusion and output projection matrices $\mathbf{W}_{\text{mr}}$ and $\mathbf{W}_{\text{out}}$. The algorithmic implementation reflects these glucose-specific adaptations through several novel components. The trend projection matrix $\mathbf{W}_{\text{trend}}$ is initialized and trained to capture glucose-physiological patterns rather than generic temporal trends. The multi-resolution convolution matrices $\mathbf{W}_{\text{conv},r}$ utilize kernel sizes corresponding to clinically relevant glucose monitoring intervals (3, 6, and 12 time steps, representing 15, 30, and 60 min windows respectively), ensuring that feature extraction aligns with diabetes management timescales. Furthermore, the feature importance weights $\mathbf{W}_{\text{imp}}$ are designed to dynamically adjust the relative contribution of glucose, basal insulin, and bolus insulin based on their temporal relevance, capturing the time-varying importance of these three parameters as they interact to influence glucose dynamics. As training progresses over multiple epochs, the weights of the model components are continuously refined, which enables the model to learn and adapt more effectively. This process provides accurate and adaptive glucose predictions $\hat{\mathbf{y}}$, offering robust performance for reliable diabetes management.

The algorithmic implementation reflects these glucose-specific adaptations through several novel components. The trend projection matrix $\mathbf{W}_{\text{trend}} \in \mathbb{R}^{D \times 3}$ is initialized and trained to capture glucose-physiological patterns rather than generic temporal trends. The multi-resolution convolution matrices $\mathbf{W}_{\text{conv},r}$ utilize kernel sizes corresponding to clinically relevant glucose monitoring intervals (3, 6, and

12 time steps, representing 15, 30, and 60 min windows respectively), ensuring that feature extraction aligns with diabetes management timescales. Furthermore, the feature importance weights $\mathbf{W}_{\text{imp}}$ are designed to dynamically adjust the relative contribution of glucose, basal insulin, and bolus insulin based on their temporal relevance, capturing the time-varying importance of these three parameters as they interact to influence glucose dynamics.

3.3. Proposed model components

3.3.1. Feature importance and time-aware embedding

The input data is initially processed in the feature importance layer, where each feature, including glucose, basal insulin, and bolus insulin, is assigned a trainable weight that is adjusted dynamically throughout the learning process. This layer employs a parameterized approach, which enables the model to learn the relative importance of each feature. As shown in Figure S1 (a), the final learned weights diverge from their uniform initialization, with the importance of CGM slightly increased while the importance of both basal and bolus insulin decreased substantially. Figure S1 (b) shows the evolution of these weights as a heatmap over the training process. CGM consistently maintains high importance, while the weights of insulin-related features decay as training progresses. Figure S1 (c) further clarifies this trend by providing a line plot that tracks the importance of each feature across epochs. These visualizations demonstrate how the model selectively emphasizes the most informative physiological signals by giving more weight to basal insulin during periods of stability while suppressing less relevant inputs. This adaptation reflects the ability of the model to discern more predictive inputs. This allows the weights to evolve as the model encounters different data scenarios. As a result, the model can prioritize features that are more predictive in certain contexts, such as giving higher importance to basal insulin during periods of stability or glucose fluctuations, while dynamically adjusting its focus in response to variations. This weighting mechanism assists the model to focus on the key physiological signals that drive glucose fluctuations, thereby reducing noise from less relevant inputs.

The weighted features are transferred into the time-aware embedding layer, which incorporates temporal context into the model [57]. A learned embedding vector is assigned to each time step, which encodes its position in time and captures essential time-based dependencies in the data. In the implementation, each time step is represented by a position encoding vector that captures both the absolute time of day (hour and minute) and the relative position within the sequence. This is achieved by a trainable embedding layer that maps each discrete

5 min interval in the 24-hour day to a dense vector. The embedding vectors are learned during model training, allowing the network to recognize patterns associated with specific times of day without requiring explicit labeling. In other words, the model can implicitly learn different glucose response patterns associated with morning periods, post-insulin administration times, and overnight intervals. This learned representation allows the model to capture temporal dependencies that are critical for accurate glucose prediction. The incorporation of this temporal information enables the model to recognize the influence of specific times on glucose trends, thereby ensuring the preservation and utilization of temporal relationships between data points for more accurate predictions.

3.3.2. Glucose trend-weighting mechanism

The glucose trend-weighting mechanism is a core component of the TW-MRT model, specifically designed to emphasize critical patterns in glucose dynamics by modeling short-term, mid-term, and long-term trends. This mechanism is designed to detect and emphasize significant glucose trends by transforming the time-embedded input sequence into multiple trend vectors: short-term, mid-term, and long-term. This transformation involves the mapping of the input sequence into trend-specific vectors, which are created using a trend projection matrix. The choice of three trend categories (short-term, mid-term, long-term) was validated through sensitivity analysis across 1–5 boundary configurations (Table S1), with the 3-boundary design achieving optimal performance. This matrix reshapes the input data to emphasize the underlying trends, which highlights key patterns relevant for prediction.

Let $\mathbf{E} \in \mathbb{R}^{N \times D}$ represent the time-embedded input sequence, where N is the sequence length, and D is the embedding dimension. $\mathbf{E} = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_D\}$ and $\mathbf{e} \in \mathbb{R}^N$ is the column vector of $\mathbf{E}$. The first step in the trend-weighting process is to compute a row vector, $\bar{\mathbf{e}} \in \mathbb{R}^{1 \times D}$, containing the mean of each column. It captures glucose pattern over the sequence and serves as the basis for projecting the input into different trend vectors. The model then applies a trend projection matrix $\mathbf{W}_{\text{trend}} \in \mathbb{R}^{D \times 3}$ as follows:

$$\mathbf{s} = \bar{\mathbf{e}} \mathbf{W}_{\text{trend}}, \quad (1)$$

where, $\mathbf{s} \in \mathbb{R}^{1 \times 3}$ represents the raw trend scores, with the three elements of $\mathbf{s}$ corresponding to the short-term, mid-term, and long-term trends, respectively. To ensure that these trend scores are interpretable as relative importance weights, the model applies the softmax function to normalize them. The softmax operation transforms the raw trend scores into probability-like weights τ_k that sum to one across the three trends:

$$\tau_k = \frac{\exp(s_k)}{\sum_{k=1}^3 \exp(s_k)}, \quad k \in \{1, 2, 3\}, \quad (2)$$

where, k represents the index for each trend component (short-term, mid-term, long-term) in the trend scores $\mathbf{s}$. These trend weights determine the relative importance of the short-term, mid-term, and long-term trends for the input sequence. To reflect these trends more effectively, the model projects the input sequence into three distinct trend-specific spaces using three learnable projection matrices $\mathbf{W}_{\text{short}}$, $\mathbf{W}_{\text{mid}}$, and $\mathbf{W}_{\text{long}}$, which correspond to short-term, mid-term, and long-term glucose patterns, respectively. These matrices capture different temporal characteristics in the glucose sequence. The final trend-weighted embedding is then computed as a weighted combination of these trend-specific projections:

$$\mathbf{E}_{\text{weighted}} = \tau_1 (\mathbf{E} \mathbf{W}_{\text{short}}) + \tau_2 (\mathbf{E} \mathbf{W}_{\text{mid}}) + \tau_3 (\mathbf{E} \mathbf{W}_{\text{long}}) \quad (3)$$

where $\mathbf{E}_{\text{weighted}} \in \mathbb{R}^{N \times D}$ is the resulting trend-weighted embedding, which dynamically adapts the input sequence based on the most relevant glucose trends at each time step. The model is now capable of focusing on patterns that represent short-term fluctuations, mid-term changes, or long-term stability, depending on the context of the data.

Algorithm 2 Trend Weighting Mechanism

- 1: **Input:** Time-embedded sequence $\mathbf{E} \in \mathbb{R}^{N \times D}$, trend projection matrix $\mathbf{W}_{\text{trend}} \in \mathbb{R}^{D \times 3}$
- 2: **Output:** Trend-weighted embeddings $\mathbf{E}_{\text{weighted}}$
- 3: Compute mean of each column: $\bar{\mathbf{e}}$
- 4: Compute raw trend scores: $\mathbf{s} = \bar{\mathbf{e}} \mathbf{W}_{\text{trend}}$
- 5: Compute trend weights: $\tau_k = \frac{\exp(s_k)}{\sum_{k=1}^3 \exp(s_k)}$, $k \in 1, 2, 3$
- 6: Compute trend-specific projections:
- 7: Short-term: $\mathbf{E}_{\text{short}} = \mathbf{E} \mathbf{W}_{\text{short}}$
- 8: Mid-term: $\mathbf{E}_{\text{mid}} = \mathbf{E} \mathbf{W}_{\text{mid}}$
- 9: Long-term: $\mathbf{E}_{\text{long}} = \mathbf{E} \mathbf{W}_{\text{long}}$
- 10: Compute weighted sum of projected embeddings: $\mathbf{E}_{\text{weighted}} = \tau_1 \mathbf{E}_{\text{short}} + \tau_2 \mathbf{E}_{\text{mid}} + \tau_3 \mathbf{E}_{\text{long}}$
- 11: **return** $\mathbf{E}_{\text{weighted}}$

The pseudocode for this mechanism is outlined in Algorithm 2, where the key steps include calculating the mean of the input embeddings across the sequence, applying the trend projection matrix, normalizing the trend scores using softmax, and finally creating the trend-weighted embedding by distributed the trend weights back to the input. The glucose trend-weighting mechanism enables the model to adaptively focus on the most relevant glucose trends at each point in time, thereby enhancing its predictive accuracy. This is achieved by emphasizing key patterns that reflect short-term, mid-term, or long-term trends in glucose levels.

3.3.3. Multi-resolution transformer

The encoded sequence is first processed by a Transformer Encoder, followed by multi-resolution convolutional fusion to integrate features across multiple time scales to comprehensively model temporal patterns in glucose dynamics. Transformer Encoder uses multi-head self-attention mechanisms to identify and capture both local patterns and global dependencies present in the entire sequence. The Transformer Encoder is crucial for understanding relationships between different time steps and features in the sequence, enabling the model to focus on important time points and patterns in parallel [4]. The self-attention mechanism calculates attention scores based on the relevance of each time step to others in the sequence, allowing the model to learn how different time points influence glucose trends. This process ensures that the model captures complex dependencies, both short-term fluctuations and long-term trends, across the entire input sequence.

After the sequence is encoded by the transformer, the model proceeds to apply multi-resolution fusion. This involves the analysis of the encoded features across different temporal scales, conducted using a series of one-dimensional convolutions with varying kernel sizes. Each convolution captures features over different time windows, extracting both rapid changes in glucose levels and slower, more sustained trends. This multi-resolution analysis ensures that the model is sensitive to patterns at different time scales, thus offering a comprehensive understanding of glucose dynamics. The outputs of these convolutions are then concatenated and fused into a single feature vector, combining information from multiple resolutions into a unified representation [47].

4. Experimental results

4.1. Dataset and performance metrics

To ensure robust validation and demonstrate generalizability across diverse populations and sensor technologies, the experimental evaluation was conducted using two datasets: OhioT1DM and D1NAMO. The OhioT1DM dataset serves as our primary evaluation benchmark, providing comprehensive, multi-parametric data for 12 Type 1 diabetes

Table 2
Hyperparameter optimization results.

HyperParameter	Tested values	Optimal value
LR	10^{-4} , 10^{-3} , 10^{-2}	10^{-3}
Dimension of the FFN	128, 256, 512	128
Number of TL	1, 2, 3	3
Number of AH	4, 8, 16	8
DR	0.1, 0.2, 0.3	0.1
WD	10^{-3} , 10^{-2} , 10^{-1}	10^{-2}

patients over an eight-week period [58]. This dataset is widely used in glucose prediction studies due to its detailed information on glucose levels, insulin doses, meal intake, and additional physiological measurements [26,29]. The glucose levels of each individual were recorded at five-minute intervals using CGM devices, thus providing a high-resolution view of glucose fluctuations. In addition to glucose levels, the dataset comprises self-reported data on daily activities, including exercise, sleep, meal consumption, and insulin doses (both basal and bolus). The OhioT1DM dataset consists of training and testing data, with approximately 9000 to 12,000 samples allocated for training and 2300 to 2900 samples for testing per patient.

To validate model generalizability and robustness across different populations and sensor technologies, we conducted additional evaluation on the D1NAMO dataset [59]. D1NAMO contains CGM data sampled every 5 min, insulin therapy records, and additional clinical variables. It includes patients using different sensor brands (e.g., Enlite, Basis) and pump models (e.g., 530G, 630G) across varied age groups, which enables a broader assessment of model adaptability. To ensure reliable evaluation on the D1NAMO dataset, we selected a subset of patients (001, 002, 006–008) who have sufficient training data and have been previously examined in published studies.

The performance of the proposed model was evaluated quantitatively using several metrics, including root mean square error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and R-squared (R^2). Additionally, the Surveillance Error Grid (SEG) was employed to assess the clinical relevance of the predictions [60]. To ensure statistical robustness, a comprehensive statistical validation was performed using paired t-tests, confidence intervals, and effect size measures. These analyses were conducted to determine the significance and magnitude of improvements. Each of these metrics provides a different perspective on the accuracy and reliability of the model. The RMSE measures the square root of the average squared differences between the predicted values and the actual values. RMSE is sensitive to large errors and provides a robust evaluation of prediction error. The MAE calculates the average absolute difference between the predicted and the actual values, providing a measure of the average prediction error. Unlike RMSE, MAE assigns equal weight to all errors, which provides a straightforward average performance of the model evaluation. The MAPE is used to express the prediction error as a percentage, which indicates the relative accuracy of the predictions. This metric is useful for understanding the magnitude of the deviations between the predicted and actual values, especially for comparisons across different scales [61]. R^2 is a statistical measure that represents the proportion of the variance for the dependent variable that is explained by the independent variables in the model. An R^2 value close to 1 indicates that the model explains a large portion of the variance, while a value close to 0 indicates poor explanatory power [62].

A Paired t-test is a statistical method used to compare two related samples or repeated measurements on the same subjects. This approach controls for subject-specific characteristics and provides more robust statistical inference than independent sample comparisons by considering individual variability through paired observations. The t-statistic is a standardized measure that quantifies the magnitude of the difference between two groups relative to the variability within the data. Larger absolute t-values provide stronger evidence against the

Table 3

Performance comparison based on the number of parameters for 30 PH in terms of RMSE (mg/dL) on OhioT1DM dataset.

Patients	1-Parameter	2-Parameter	3-Parameter	4-Parameter
540	15.35	12.78	12.97	13.32
544	9.96	9.76	9.67	9.85
552	7.58	7.63	7.74	7.82
559	10.85	10.67	10.52	10.63
563	11.59	11.48	11.54	11.82
567	11.17	11.11	11.22	11.11
570	9.55	9.81	9.70	9.65
575	14.41	14.34	14.58	14.58
584	12.68	12.97	12.81	12.82
588	11.04	11.49	11.39	11.34
591	12.98	13.59	13.41	13.33
596	10.48	11.09	10.93	11.32
Average	11.47	11.39	11.37	11.47

null hypothesis of no difference. P-values represent the probability of observing a test statistic as extreme as or more extreme than the one calculated, assuming the null hypothesis is true. A p -value less than 0.05 is conventionally considered statistically significant, indicating strong evidence against the null hypothesis. 95% confidence intervals provide a range of values that likely contains the true population parameter. For mean differences, confidence intervals offer insight into the precision of the estimate and the magnitude of the effect. Wide intervals indicate inaccurate predictions, while small intervals suggest precise predictions. Cohen's d is a standardized measure of effect size that quantifies the magnitude of the difference between two groups in terms of standard deviation units. Effect sizes are typically interpreted as small ($d \approx 0.2$), medium ($d \approx 0.5$), or large ($d \approx 0.8$) with values exceeding 0.8 indicating a very large effect. Effect size measures complement statistical significance testing by indicating whether observed differences are practically meaningful, as statistical significance can be achieved with trivial differences given a large enough sample size.

On the other hand, SEG was used to evaluate the clinical accuracy of the model. The SEG is a specifically designed tool for CGM that categorizes prediction errors based on their potential impact on clinical decision-making [63]. The SEG classifies the predicted glucose values into zones that reflect the probability of those values resulting in the appropriate or inappropriate implementation of treatment actions. This provides a visual and quantitative understanding of the clinical relevance of the prediction errors.

Patients were modeled independently using their own data, with a separate model trained for each case. However, all models were trained using the same preprocessing steps, architecture, and hyperparameter settings. The training process was performed uniformly across all patients without introducing individual-specific fine-tuning or manual modifications. This approach ensured a consistent training pipeline across individuals, allowing the model to capture patient-specific glucose dynamics while maintaining methodological consistency and comparability.

4.2. Hyperparameter tuning

The selection of hyperparameters is a critical step in determining the performance of DL models. In order to achieve optimal prediction results for each PH, a range of hyperparameters was tested, and the configurations were selected according to the lowest error. The key hyperparameters in this study included the learning rate (LR), the dimension of the feedforward network (FFN), the number of transformer layers (TL), the number of attention heads (AH), the dropout rate (DR), and the weight decay (WD). The final configuration, shown in Table 2, includes a LR of 10^{-3} , a FFN dimension of 128, three TL, eight AH, a dropout rate of 0.1, and a WD of 10^{-2} . Furthermore, the input window sizes of 3, 6, 9, 12, 18, and 24-time steps were tested for each PH of 15, 30, 45, 60, 90, and 120 min. The most effective input window size

Table 4
Prediction performance at 15, 30, 45 PHs for each patient on OhioT1DM dataset.

Patients	15 PH				30 PH				45 PH			
	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2
540	7.35	4.24	2.80	0.99	12.97	8.18	5.52	0.96	19.37	12.39	8.39	0.91
544	5.42	3.31	2.24	0.99	9.67	5.91	3.98	0.96	13.50	8.22	5.52	0.93
552	4.06	2.50	1.94	0.99	7.74	4.47	3.35	0.97	11.32	6.63	4.94	0.93
559	5.62	3.31	2.12	0.99	10.52	6.23	4.05	0.97	15.16	9.15	6.02	0.94
563	7.30	4.19	2.65	0.97	11.54	6.91	4.37	0.93	15.50	9.64	6.04	0.88
567	5.58	3.42	2.45	0.99	11.22	6.84	5.01	0.96	16.43	10.25	7.52	0.92
570	5.38	3.41	1.66	0.99	9.70	6.15	3.03	0.98	13.62	8.75	4.33	0.96
575	8.25	4.17	2.90	0.98	14.58	7.86	5.44	0.94	19.02	11.31	8.12	0.90
584	7.00	4.26	2.87	0.99	12.81	7.85	5.21	0.96	17.77	11.14	7.33	0.91
588	6.14	4.03	2.45	0.98	11.39	7.46	4.48	0.94	15.95	10.52	6.34	0.89
591	7.20	4.38	3.40	0.98	13.41	8.96	7.23	0.93	17.83	11.74	9.39	0.88
596	6.19	3.70	2.79	0.98	10.93	6.70	4.99	0.95	14.81	9.53	7.11	0.91
Average	6.29	3.74	2.52	0.99	11.37	6.96	4.72	0.96	15.86	9.94	6.76	0.91

was 3-time steps, corresponding to 15 min intervals, which allowed the model to best capture glucose fluctuations. The model was trained for a maximum of 200 epochs, with each epoch taking approximately 5 s, and the best-performing epoch selected based on the lowest loss value. Furthermore, we applied the preprocessing pipeline to the OhioT1DM dataset for patients with the following IDs: 540, 544, 552, 559, 563, 567, 570, 575, 584, 588, 591, and 596. This systematic approach ensured the quality of the data and the performance of the model. Missing values in the CGM data were handled using linear interpolation, followed by backward and forward filling for any remaining gaps. UKS was implemented using Merwe scaled sigma points with the following optimized parameters: $\alpha = 0.001$ (sigma point spread), $\beta = 2.0$ (Gaussian distribution parameter), $\kappa = 0$ (scaling factor), measurement noise variance $R = 0.1$, process noise variance $Q = 0.1$, and initial covariance scaling $P \times 10$. This smoothing process effectively reduced noise while preserving glucose trend fidelity. Feature normalization was performed using StandardScaler to ensure a zero mean and unit variance across all input features. The dataset was split into training and validation sets using a 80/20 ratio with sequential (non-shuffled) partitioning to preserve the temporal order that is essential for time series prediction.

The TW-MRT model demonstrates exceptional computational efficiency with 462,346 total parameters and a compact 1.76 MB model size, achieving optimal balance between predictive accuracy and computational practicality. To contextualize this efficiency, our model is 240 times smaller than BERT-base (110 million parameters) and 378,000 times smaller than GPT-3 (175 billion parameters), while maintaining transformer capabilities for medical prediction. The analysis of inference performance reveals a time of 12.9 ms per prediction and a rate of 77.5 predictions per second. The model requires only 1.84 MB of memory for inference, whereas BERT-base requires approximately 440 MB and GPT-2 requires 468 MB. This 238-fold reduction in memory usage enables deployment on resource-constrained medical devices, which would be difficult with traditional transformer models.

The experiments were carried out using two identical hardware platforms, both equipped with an Intel(R) Core i9-9940X processor and an NVIDIA GeForce GTX 2080 Ti GPU. The software environment consisted of Python 3.9 and Pytorch 2.12.0. To ensure reproducibility, the details of the computational setup, including hardware configurations, and software frameworks, are provided in Table S2. The source code, implementation details, and instructions for reproducing the experiments in this study are publicly available at <https://github.com/atilimkoca/glucose-prediction-TW-MRT>.

4.3. Results and discussion

An analysis is conducted on the performance of the proposed TW-MRT model, with a particular focus on its predictive errors across a range of PH. The model was evaluated based on the results of comprehensive experimentation utilizing key performance metrics, including

RMSE, MAE, MAPE, and R^2 , to measure its predictive capabilities. The analysis was performed using a comparison of parameter configurations, an ablation study to highlight the contributions of each components, and a clinical relevance assessment using SEG. Furthermore, the performance of the model was compared with state-of-the-art methods in glucose prediction.

Table 3 compares performance of the model based on parameters used for 30 min PHs. The results demonstrate that the three-parameter model exhibited the most optimal performance, displaying the lowest RMSE of 11.37 mg/dL, which was superior to that of the two-parameter model (11.39 mg/dL), the four-parameter model (11.47 mg/dL), and the one-parameter model (11.47 mg/dL). The selection of these parameter combinations was determined based on the correlations between the parameters in the dataset and glucose values, as illustrated in the correlation heatmap (Figure S2). The one-parameter model incorporates only glucose values from CGM, whereas the two-parameter model includes basal insulin due to its moderate correlation with CGM. The three-parameter model incorporates both basal and bolus insulin, which are important factors in glucose dynamics. The four-parameter model incorporates carbohydrate intake (CHO), which has a relatively weak correlation with glucose. This has resulted in an increased margin of error and added additional complexity, possibly contributing to overfitting. Given this observation, the evaluation process was limited with four parameters to avoid model degradation and focus on features that consistently showed strong predictive value. The incorporation of additional parameters beyond the three key parameters has limited potential to improve model performance.

The model was evaluated across a range of PHs, between 15 and 120 min. As illustrated in Table 4, the model demonstrated robust performance for shorter time horizons. The mean RMSE for the 15 min horizon was 6.29 mg/dL, while for the 30- and 45 min horizons, the RMSE exhibited a slight increase to 11.38 mg/dL and 15.86 mg/dL, respectively. As the PH increased to 60, 90, and 120 min, the RMSE exhibited a further increase, reflecting the difficulty in accurately predicting glucose levels over extended periods, as illustrated in Table 5. However, the model still maintained reasonably accurate predictions, with an RMSE of 19.91 mg/dL for the 60 min horizon and 31.68 mg/dL for the 120 min horizon. The R^2 values remained high for shorter horizons, indicating that the model was able to explain a large portion of the variance in the data.

The clinical impact of the model was evaluated using the SEG and Fig. 3 illustrates the SEG results for patients 552 and 575 at the 30 min PH. The patients were selected to represent the range of model performance, as detailed in Table 4. Patient 552 exhibited the lowest RMSE (7.75 mg/dL), while patient 575 exhibited the highest RMSE (14.59 mg/dL). Despite these differences, the majority of predictions for both patients exhibited a green rating on the SEG, indicating that the model is clinically reliable even in the worst-case scenario. This suggests that the model may maintain clinically acceptable accuracy across patients with varying prediction errors, highlighting its potential

Table 5

Prediction performance at 60, 90, 120 PHs for each patient on OhioT1DM dataset.

Patients	60 PH				90 PH				120 PH			
	RMSE	MAE	MAPE	R ²	RMSE	MAE	MAPE	R ²	RMSE	MAE	MAPE	R ²
540	24.71	16.07	10.98	0.86	33.25	22.25	15.39	0.74	39.37	26.80	18.58	0.64
544	17.12	10.53	7.13	0.89	22.90	14.72	10.10	0.79	27.61	18.58	12.94	0.70
552	14.42	8.10	6.12	0.88	19.47	10.90	8.36	0.78	23.07	13.33	10.28	0.69
559	19.29	12.00	8.01	0.91	26.61	17.16	11.56	0.83	32.60	21.71	14.73	0.74
563	19.30	12.33	7.68	0.82	24.70	16.14	10.14	0.70	29.04	19.61	12.37	0.59
567	21.22	13.63	10.17	0.87	28.49	18.90	14.27	0.76	34.59	24.12	19.08	0.64
570	17.21	11.13	5.55	0.93	24.02	16.20	8.00	0.87	29.48	20.27	10.18	0.81
575	23.20	14.77	10.90	0.85	29.79	19.89	14.88	0.76	34.96	24.01	17.93	0.67
584	22.24	14.28	9.33	0.86	29.86	19.87	13.17	0.75	35.26	24.11	16.12	0.65
588	19.50	13.10	8.04	0.83	25.63	17.42	10.57	0.71	30.74	21.33	13.03	0.58
591	22.02	14.73	11.81	0.82	28.51	19.85	16.30	0.69	33.18	23.86	20.03	0.59
596	18.74	12.49	9.33	0.86	25.40	17.06	12.61	0.74	30.31	20.71	15.30	0.62
Average	19.91	12.76	8.75	0.86	26.55	17.53	12.11	0.76	31.68	21.54	15.05	0.66

robustness. Therefore, the reliability of real-time glucose monitoring systems is crucial, as inaccurate predictions have the potential to result in inappropriate medical interventions. To strengthen the clinical relevance, we included SEG plots for all evaluated patients in the supplementary material (Figure S3). We extended the SEG analysis to the D1NAMO dataset. Figure S4 presents the plots for patients 1, 2, 5, 6, 7, and 8. These evaluations suggest that the model may perform acceptably across different datasets and patient profiles. However, further validation is necessary to support broader generalization.

Model interpretability was examined through visualization of attention weights and analysis of feature importance based on SHAP. Figure S5 (a) illustrates the temporal focus of the model and reveals that recent timesteps, particularly $t-0$, receive the highest attention weights. This suggests that the model places greater emphasis on the most recent glucose values during prediction, which aligns with physiological expectations. SHAP analysis (Figure S5 (b)) quantifies the contribution of each input feature, complementing this finding. CGM values emerged as the dominant factor influencing predictions (mean SHAP value: 1.6208), while other variables, such as basal insulin, bolus insulin, and carbohydrate intake, contributed minimally. These results are consistent with the parameter selection findings and confirm that the model relies on clinically meaningful and physiologically relevant inputs to generate accurate prediction.

To assess clinical deployment readiness, comprehensive inter-patient variability analysis was conducted across all 12 patients in the OhioT1DM dataset. The TW-MRT model demonstrated robust performance consistency with a coefficient of variation of 16.5% (mean RMSE: 11.37 ± 1.88 mg/dL, range: 7.74–14.58 mg/dL), indicating manageable variability within medical device acceptance criteria. Statistical analysis revealed minimal performance differences across patient demographics, technology specifications, or temporal cohorts (Table S3), with all p -values > 0.05 , demonstrating strong generalizability across gender, age groups, pump models (630G vs 530G), sensor types (Enlite vs Basis), and data collection periods (2018 vs 2020). Age group comparison between younger (20–40 years: 10.41 ± 2.62 mg/dL) and older (40–60 years: 11.69 ± 1.64 mg/dL) participants demonstrated consistent performance across age ranges ($t = -1.027$, $p = 0.329$). The younger age group showed greater performance variability ($SD = 2.62$ mg/dL), potentially reflecting more diverse lifestyle patterns, while the older group demonstrated more consistent performance ($SD = 1.64$ mg/dL). Despite these variability differences, the lack of statistical significance confirms the ability of the model to maintain reliable performance across different age demographics. These findings confirm the suitability of the model for clinical deployment with technology-agnostic performance and clear risk-stratified monitoring protocols.

Furthermore, we conducted a comparison against a simple attention-based weighting baseline (Table S4) to demonstrate the incremental benefits of our proposed trend-weighting mechanism over traditional attention mechanisms. This analysis directly addresses whether our

physiologically motivated trend-weighting provides meaningful improvements over standard attention mechanisms. The results clearly show the superiority of the trend-weighting mechanism, which achieved an average RMSE of 11.37 mg/dL, compared to 12.21 mg/dL for the simple attention baseline (+0.84 mg/dL improvement; 6.9% reduction in prediction error). Similarly, mean absolute error (MAE) improved from 7.76 mg/dL to 6.96 mg/dL, representing a 10.3% enhancement in prediction accuracy. Patient-level analysis reveals consistent improvements across all 12 patients, with the most substantial gains observed in patients with higher glucose variability (e.g., patient 591, with an RMSE improvement from 16.18 to 13.41 mg/dL). This consistency across patients demonstrates that the trend-weighting mechanism provides robust benefits across diverse glucose dynamics patterns. It confirms that our physiologically grounded approach to temporal weighting offers advantages over traditional attention mechanisms for glucose prediction tasks.

The performance of the proposed TW-MRT model was compared with several state-of-the-art models, as shown in Table 6. In order to ensure a fair comparison, we selected studies utilizing the OhioT1DM dataset and reporting RMSE and MAE metrics for similar PH values. The proposed model demonstrates superior performance compared to all baseline models across the evaluated PHs. For the 30 min PH, the proposed model achieves an RMSE of 11.37 mg/dL and an MAE of 6.96 mg/dL, which represents a notable improvement over alternative models such as the multi-head attention network (RMSE of 20.57 mg/dL) and the ensemble neural network-based models (RMSE of 19.57 mg/dL). Despite its strength in capturing local patterns through focused attention mechanisms, the multi-head attention network model encounters difficulties in maintaining consistent accuracy across varying time scales due to its tendency to emphasize global patterns over localized trends. On the other hand, ensemble neural network models, which combine predictions from multiple individual models, can suffer from increased computational complexity and difficulty in balancing the weights of different models, resulting in less consistent performance.

The weighted decision level fusion achieves an RMSE of 21.90 mg/dL. Although this model employs multiple deep neural networks with fusion strategies, it encounters challenges in distributing optimal weights across different prediction components [26]. The autonomous channel deep learning framework (RMSE: 18.93 mg/dL) processes multi-lag inputs. However, it requires computational resources to coordinate multiple asynchronous channels [29]. The GAN with GRU model demonstrates competitive performance (RMSE: 18.37 mg/dL) with adversarial training mechanisms. However, its generative framework introduces training instability and convergence challenges [30]. The neural physiological encoder (RMSE: 17.80 mg/dL) uses decomposed convolutional filters to process physiological signals, but it lacks the sophisticated temporal modeling capabilities necessary for complex glucose dynamics [32]. The feature transformation Bi-LSTM model shows promising results (RMSE: 16.81 mg/dL) through bidirectional processing and feature engineering. However, its sequential nature

Table 6

The performance comparison with state-of-the-art models in terms of RMSE and MAE metrics (mg/dL) on OhioT1DM dataset.

Method	30 PH		60 PH	
	RMSE	MAE	RMSE	MAE
Weighted decision level fusion [26]	21.90	15.87	35.10	26.41
Multi-head attention network [38]	20.57	–	35.09	–
Ensemble neural network [27]	19.57	14.06	34.93	25.95
Evidential DL with meta-learning [37]	19.46	13.25	31.07	22.86
GluNet [28]	19.28	–	31.83	–
Edge-LSTM [35]	19.10	13.59	32.61	24.25
Population-Specific TFT [44]	19.10	13.10	32.30	23.20
Edge-based temporal fusion transformer [43]	19.09	13.07	32.31	23.24
Autonomous channel DL [29]	18.93	–	31.79	–
DL with IoMT and RNN [25]	18.92	13.46	32.54	24.05
DRNNs [34]	18.60	–	–	–
GAN with GRU [30]	18.37	13.1	32.21	23.5
DRL using LSTM [31]	18.32	12.93	–	–
Neural physiological encoder [32]	17.80	–	–	–
Feature transformation Bi-LSTM [3]	16.81	10.8	25.48	18.6
Stacked LSTM with multitask [24]	16.06	10.64	30.89	22.07
Crossformer [42]	15.81	9.67	36.12	16.38
Attention-based multilayer GRU [36]	13.11	8.35	28.25	14.87
GLIMMER [33]	–	–	23.97	15.83
Proposed model	11.37	6.96	19.91	12.76

limits the efficiency of parallelization [3]. For longer-term predictions, GLIMMER achieves an RMSE of 23.97 mg/dL for 60 min predictions through modified error rate calculations. However, its CNN-LSTM hybrid architecture is unable to adequately capture the multi-scale temporal dependencies essential for extended PH [33].

The evidential DL model, which offers uncertainty measures, encounters challenges with data deficiency, particularly during the initial training phase, which can result in a slower adaptation to new patients. Furthermore, its emphasis on meta-learning and RNN attention mechanisms can prove challenging in balancing short and long-term dependencies [37]. In comparison, the proposed TW-MRT model demonstrates superior performance in handling these complexities, achieving a lower RMSE of 11.37 mg/dL for the 30 min PH. Similarly, the edge-LSTM model, which produced an RMSE of 19.10 mg/dL for the same time horizon, also demonstrates higher error rates compared to the proposed model, which highlights the limitations of LSTM-based models in capturing temporal relationships in glucose dynamics [35]. The dilated recurrent neural network (DRNN) and the deep reinforcement learning (DRL) model using an LSTM represent notable advancements in addressing temporal complexity in glucose prediction [34]. The DRNN model achieves an RMSE of 18.60 mg/dL for 30 min predictions by using dilated convolutions to increase the size of the receptive field and capture long-term dependencies. The DRL using LSTM model demonstrates competitive short-term performance with an RMSE of 18.32 mg/dL and MAE of 12.93 mg/dL for 30 min predictions, employing reinforcement learning principles to optimize LSTM parameters [31]. Despite the sophisticated training methodology, this model still exhibits a 61.1% higher RMSE compared to the proposed TW-MRT model. Recent developments in transformer-based glucose prediction have produced three models that currently define state-of-the-art performance benchmarks. The crossformer model achieves an RMSE of 15.81 mg/dL for 30 min predictions through cross-dimensional attention mechanisms and patch-wise processing [42]. However, this model shows significant inconsistency in performance across different PH, with its RMSE increasing to 36.12 mg/dL for 60 PH. This indicates limited robustness for extended prediction. The edge-based temporal fusion transformer (E-TFT) demonstrates competitive performance with RMSE values of 19.09 mg/dL and 32.31 mg/dL for 30 and 60 min predictions, respectively [43]. While this model successfully implements transformer architecture on edge computing devices, it relies on generic temporal fusion mechanisms without glucose-specific physiological adaptations. The population-specific TFT achieves similar performance with RMSE values of 19.10 mg/dL and 32.30 mg/dL for 30

and 60 min horizons, incorporating demographic data for personalization [44]. However, this model focuses on population-level adaptation rather than the underlying glucose–insulin physiological dynamics that drive prediction accuracy.

A comparison of the proposed model with our previous study [36], Attention Based Multilayer GRU EnDe, which reported an RMSE of 13.11 mg/dL for the 30 min PH, demonstrates a 13.2% improvement, with the RMSE reduced to 11.37 mg/dL. This considerable enhancement is largely attributable to the incorporation of a multi-resolution transformer and trend-weighting mechanism, which facilitate more precise modeling of glucose patterns in comparison to the GRU-based architecture employed in the previous study.

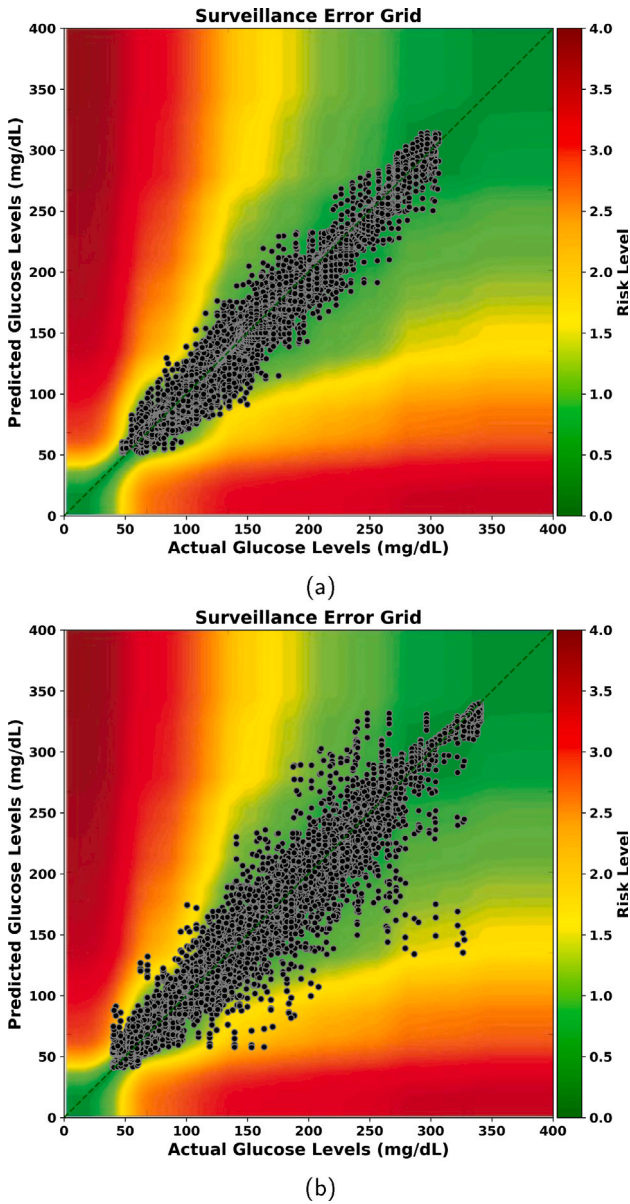
To statistically validate and ensure the robustness of the observed improvements, we conducted paired t-tests against multiple baseline models. This analysis used studies that provided individual, patient-specific results for all 12 patients in the OhioT1DM dataset. This enabled proper paired comparisons that considered patient-specific glucose dynamics. Table 3 lists the per-patient RMSE values of the three-parameter TW-MRT configuration, which served as the reference in all paired tests. As shown in Table S5, the statistical validation demonstrates that the proposed TW-MRT model achieves significant improvements across all baseline comparisons. The paired t-test results show strong statistical evidence, with p-values ranging from 6.05×10^{-4} to 7.22×10^{-11} . The t-statistics range from 4.74 to 24.08, demonstrating substantial differences relative to the variability in the data. The mean improvements are clinically substantial, ranging from 1.73 mg/dL with the Attention-based Multilayer GRU to 10.53 mg/dL with the Weighted Decision Level Fusion model. The 95% confidence intervals are all narrow and positive, ranging from 1.31 to 2.16 and from 9.57 to 11.49. This confirms the precision and reliability of the observed performance gains. Notably, the effect sizes (Cohen's d) range from 1.37 to 6.95 and are all categorized as “very large”, indicating exceptional practical significance in addition to statistical significance. The largest effect size, 6.95, was observed against the Weighted Decision Level Fusion model and represents a remarkably large improvement, exceeding conventional thresholds for meaningful clinical differences by a significant margin. The consistency of the statistical results obtained from multiple baseline models, combined with the significant effect sizes, suggests that the improvements achieved with the proposed TW-MRT model represent significant advancements in glucose prediction accuracy, regarding to its new architectural components and training methodology.

To further validate the generalizability and robustness of the proposed TW-MRT model across diverse populations and sensor technologies, we conducted comprehensive cross-dataset evaluation on the

Table 7

Performance Comparison of TW-MRT and State-of-the-Art Time Series Models on D1NAMO Dataset.

Patient ID	Proposed model		Crossformer [64]		HDMixer [65]		Informer [66]		Fedformer [67]		Pathformer [68]	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
001	15.97	10.96	24.60	17.20	17.64	12.66	35.46	25.48	29.72	21.51	13.68	9.17
002	14.48	8.11	35.85	26.66	24.87	17.43	47.59	33.96	35.11	27.24	15.59	10.02
005	14.99	10.82	20.50	16.68	17.72	14.13	35.06	24.41	25.31	18.59	11.67	7.43
006	5.10	3.80	14.75	11.12	5.95	4.59	27.45	22.71	19.19	13.74	8.36	5.07
007	7.63	5.58	14.04	11.37	15.19	11.82	23.86	17.75	12.69	9.08	7.09	4.59
008	9.61	7.82	19.72	14.22	10.12	7.33	29.89	25.76	23.94	16.90	12.82	8.57
Average	11.30	7.85	21.58	16.24	15.25	11.33	32.39	25.01	24.31	17.84	11.54	7.48

**Fig. 3.** SEG of patient 552 (a) and 575 (b) for 30 PH.

D1NAMO dataset. This validation is crucial as it demonstrates model performance beyond the primary OhioT1DM dataset, addressing potential concerns about overfitting to specific demographic characteristics or sensor technologies.

For cross-dataset validation, we evaluated TW-MRT against multiple categories of baseline models to ensure comprehensive assessment.

First, we compared against state-of-the-art time series prediction models including Crossformer [64], HDMixer [65], Informer [66], FEDformer [67], and Pathformer [68]. These models represent recent advances in transformer-based time series prediction and provide strong baselines for temporal pattern modeling. Additionally, we compared against previously published diabetes-specific models that have been evaluated on D1NAMO, including BAGGING metaR [18] and Bayesian Ridge regression [19], enabling direct performance comparisons with established glucose prediction baselines.

The cross-dataset validation results demonstrate consistent superior performance of TW-MRT across all baseline comparisons (Table 7). Against state-of-the-art time series models, TW-MRT achieved substantial improvements in both RMSE and MAE metrics across all six D1NAMO patients (001, 002, 005, 006, 007, 008). Notably, TW-MRT outperformed Crossformer with an average RMSE improvement of 10.28 mg/dL (21.58 vs 11.30 mg/dL), representing a 47.6% reduction in prediction error. Similarly, significant improvements were observed against HDMixer (4.02 mg/dL improvement), Informer (21.09 mg/dL improvement), FEDformer (13.01 mg/dL improvement), and Pathformer (0.24 mg/dL improvement). The consistent performance gains across diverse architectural model—from attention-based transformers to hierarchical dependency models—demonstrate the robustness of trend-weighting mechanism and multi-resolution transformer design.

Furthermore, comparisons with diabetes-specific baselines previously evaluated on D1NAMO provide additional validation evidence (Tables S6–S7). Against BAGGING metaR using the same patient subset (001, 006, 007, 008), TW-MRT achieved a mean RMSE of 9.58 mg/dL compared to 17.66 mg/dL, representing a 45.7% improvement. TW-MRT demonstrated a 63.5% improvement with a mean RMSE of 0.58 mmol/L over Bayesian Ridge regression on patients 002, 007, and 008 using mmol/L units. These results are particularly significant as they represent direct comparisons with previously published diabetes prediction models using identical patient data, ensuring fair and unbiased evaluation.

In conclusion, the proposed TW-MRT model demonstrates consistent superiority over existing state-of-the-art methods across all PHs, with notable improvements in error. The integration of multi-resolution fusion and trend-weighted mechanisms enables the model to capture complex glucose dynamics superior to existing models, resulting in reduced prediction errors across both short and long-term horizons. Furthermore, the average inference time per prediction using the TW-MRT model was 0.0829 ms on an NVIDIA RTX 2080 Ti GPU, demonstrating its efficiency for real-time glucose prediction.

4.4. Ablation study

In order to validate the effectiveness of the proposed TW-MRT model, a series of ablation experiments were conducted to determine the contribution of each core component. As detailed in Table 8, these experiments involved removing key components from the model architecture, ranging from single-component ablations to lightweight configurations optimized for mobile deployment, and subsequently analyzing the resulting impact on performance for the 30 min PH.

Table 8
Comprehensive ablation study results for 30 min PH on OhioT1DM dataset.

Model configuration	RMSE (mg/dL)	Std Dev	Parameters
Proposed Model (Full)	11.37	-	92,233
<i>Single Component Ablations:</i>			
Without Time-Aware Embedding	12.65	1.63	92,041
Without Feature Importance	12.80	1.83	51,017
Without Multi-Resolution Fusion	12.94	1.81	50,825
Without Glucose Trend-Weighting	12.81	1.94	92,038
Without UKS	12.93	1.86	51,020
<i>Dual Component Ablations:</i>			
Without Time-Aware + Feature Importance	12.74	1.66	92,041
Without Time-Aware + Multi-Resolution	12.80	1.78	50,828
Without Time-Aware + Glucose Trend	12.59	1.63	91,849
Without Feature Importance + Multi-Resolution	12.94	1.80	51,017
Without Multi-Resolution + Glucose Trend	12.72	1.89	50,825
<i>Triple Component Ablations:</i>			
Without Time-Aware + Feature + Multi-Res	12.86	1.73	50,825
Without Feature + Multi-Res + Glucose Trend	12.67	1.84	50,822
<i>Baseline Comparisons:</i>			
Basic Transformer Only	12.88	1.66	50,630
Lightweight Model (Mobile)	12.67	1.84	50,822

The full model, which incorporates all of the proposed components, demonstrated the lowest RMSE of 11.37 mg/dL with 92,233 parameters. The systematic removal of specific components revealed both individual component importance and practical deployment strategies.

Single Component Analysis: Individual component removal revealed the following hierarchy of importance. The exclusion of the multi-resolution fusion layer resulted in the highest performance degradation, with RMSE reaching 12.94 mg/dL (+1.57 mg/dL), demonstrating that integrating data from different time scales is the most critical component for predictive accuracy. The removal of the glucose trend-weighting mechanism resulted in an RMSE of 12.81 mg/dL (+1.44 mg/dL), highlighting the importance of capturing both current and historical trends in glucose fluctuations. Similarly, the removal of feature importance resulted in an RMSE of 12.80 mg/dL (+1.43 mg/dL), indicating that the capacity of model to assess feature relevance has substantial impact on prediction accuracy. The exclusion of the time-aware embedding layer resulted in an RMSE of 12.65 mg/dL (+1.28 mg/dL), emphasizing the significance of temporal alignment in capturing dynamic glucose patterns. Lastly, the removal of UKS resulted in an RMSE of 12.93 mg/dL (+1.56 mg/dL), indicating that the smoothing effect provided by UKS helps reduce noise in the data, leading to more stable predictions. To validate the choice of UKS for noise reduction, we compared it against alternative smoothing techniques including exponential smoothing, Savitzky–Golay filtering, moving average, and wavelet filtering (Table S8). UKS demonstrated superior performance with the lowest RMSE (11.37 mg/dL) and MAE (6.96 mg/dL), outperforming Savitzky–Golay (12.14 RMSE), wavelet filtering (12.14 RMSE), moving average (12.03 RMSE), and exponential smoothing (11.85 RMSE). This validates UKS as the optimal preprocessing techniques for our glucose prediction task. **Practical Deployment Configurations:** To address computational efficiency requirements, we evaluated dual and triple component ablations that provide intermediate complexity options. The dual component ablations achieved RMSE values ranging from 12.59 to 12.94 mg/dL with varying parameter counts, offering flexibility for different deployment scenarios. Most importantly, our lightweight model configuration, achieved by removing multi-resolution fusion, glucose trend-weighting, and feature importance components, demonstrates optimal practicality with only 12.67 mg/dL RMSE (+1.30 mg/dL degradation) while reducing parameters to 50,822 (45% reduction). This lightweight configuration maintains clinically acceptable performance while enabling deployment on resource-constrained mobile and edge devices. **Computational Efficiency Analysis:** The parameter efficiency analysis reveals strategic

options for different deployment scenarios. For maximum accuracy in clinical settings with sufficient computational resources, the full model provides optimal performance. For mobile health applications requiring real-time processing, the lightweight model offers an excellent balance of accuracy and efficiency. The basic transformer-only configuration (12.88 mg/dL RMSE, 50,630 parameters) provides a minimal baseline, demonstrating that our proposed components collectively contribute significant performance improvements over standard transformer architectures. Overall, these comprehensive experiments confirm that each component plays a critical role in enhancing the predictive power of the proposed model, while also providing practical deployment strategies. The combined contributions of these components enable the TW-MRT model to effectively capture complex glucose dynamics and provide optimal performance across different computational constraints, from high-accuracy clinical systems to efficient mobile applications.

5. Conclusion

In this study, we proposed a TW-MRT model for multi-parametric glucose prediction, designed to address the limitations of traditional AI methods in handling the complex, non-stationary nature of CGM data. Unlike conventional models that depend on single-scale analysis or static modulation, the TW-MRT model integrates a trend-weighting mechanism with multi-resolution analysis, enabling it to capture both short- and long-term dependencies in glucose dynamics more effectively. Additionally, the multi-parametric focus of the model allows it to process and learn from multiple input parameters simultaneously, enhancing its ability to model the intricate interactions between glucose and insulin. This multi-dimensional approach provides a more comprehensive and adaptive solution for accurate glucose prediction.

Extensive experimental evaluations on the OhioT1DM dataset demonstrate the superiority of the proposed model across multiple PHs, including 15, 30, 45, 60, 90, and 120 min. The TW-MRT consistently outperforms state-of-the-art methods, achieving RMSE values of 6.29 mg/dL, 11.38 mg/dL, 15.86 mg/dL, 19.91 mg/dL, 26.55 mg/dL, and 31.68 mg/dL for the respective horizons. Furthermore, the SEG analysis confirms the clinical reliability of the predictions, even for patients with higher variability in glucose levels.

Furthermore, the ablation study demonstrates the importance of each component in the proposed model, indicating that the combination of trend-weighting mechanism and multi-resolution fusion is a principal factor in the predictive success of the model. To the best of our knowledge, this is the first study to integrate trend-weighting

mechanisms within a transformer-based model for glucose prediction, highlighting its potential for enhancing diabetes management.

In future work, the TW-MRT model could be extended to include self-learning capabilities, allowing it to adapt to individual user data and refine its predictions over time. Additionally, integrating multi-modal data from wearable sensors — such as sleep patterns, physical activity, and stress levels — may further enhance prediction accuracy and personalization. Moreover, the integration of this model into a real-time monitoring system, potentially within a closed-loop artificial pancreas framework, presents an opportunity to further advance personalized diabetes care.

CRedit authorship contribution statement

Ömer Atılım Koca: Writing – original draft, Software, Investigation. **Volkan Kılıç:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.bspc.2025.108885>.

Data availability

The source code, implementation details, and instructions for reproducing the experiments in this study are publicly available in github link provided in manuscript.

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